



Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA	Div: B	Course Code: MCA01605	Batch: S1
Semester: III		Course Name: Web Programmin Laboratory	
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CO No: CO606.4		Assignment No: 12	

Title: Write a program to implement web service for calculator applications using PHP.

Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>PHP Web Service Calculator</title>
</head>
<body>
  <div class="container">
    <h1>Calculator</h1>
    <p class="subtitle">Using a PHP Web Service</p>
    <form id="calculatorForm">
      <div>
        <label for="num1">First Number</label>
        <input type="number" id="num1" name="num1" step="any" required>
      </div>
      <div>
        <label for="num2">Second Number</label>
        <input type="number" id="num2" name="num2" step="any" required>
      </div>
      <div class="button-row">
        <button data-op="add" class="op-btn">+</button>
        <button data-op="subtract" class="op-btn">-</button>
        <button data-op="multiply" class="op-btn">&times;</button>
        <button data-op="divide" class="op-btn">&divide;</button>
      </div>
    </form>
    <div id="result-container">
      <p id="result-text">Result will be shown here</p>
    </div>
  </div>
```



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</div>

<script>

```
document.addEventListener('DOMContentLoaded', () => {  
    const form = document.getElementById('calculatorForm');  
    const resultText = document.getElementById('result-text');  
    const resultContainer = document.getElementById('result-container');  
    const operationButtons = document.querySelectorAll('.op-btn');  
    const webServiceUrl = 'calculator_ws.php';
```

```
    operationButtons.forEach(button => {  
        button.addEventListener('click', async (event) => {  
            event.preventDefault();  
            const operation = event.target.dataset.op;  
            const formData = new FormData(form);  
            formData.append('operation', operation);  
  
            if (!formData.get('num1') || !formData.get('num2')) {  
                displayResult('Please enter both numbers.', true);  
                return;  
            }  
        })  
    })
```

```
    try {  
        const response = await fetch(webServiceUrl, {  
            method: 'POST',  
            body: formData  
        });  
        if (!response.ok) {  
            throw new Error(`HTTP error! status: ${response.status}`);  
        }  
        const data = await response.json();  
        if (data.error) {  
            displayResult(data.message, true);  
        } else {  
            displayResult(`Result: ${data.result}`);  
        }  
    } catch (error) {  
        console.error('Fetch Error:', error);  
        displayResult('Error communicating with the server.', true);  
    }  
});  
});
```

```
function displayResult(message, isError = false) {
```



```
resultText.textContent = message;
if (isError) {
    resultContainer.classList.add('error');
} else {
    resultContainer.classList.remove('error');
}
}
});
</script>
</body>
</html>
```

calculator_ws.php

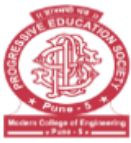
```
<?php
header('Content-Type: application/json');

$response = ['error' => false, 'result' => 0, 'message' => ''];

if (isset($_POST['num1'], $_POST['num2'], $_POST['operation'])) {
    $num1 = $_POST['num1'];
    $num2 = $_POST['num2'];
    $operation = $_POST['operation'];

    if (!is_numeric($num1) || !is_numeric($num2)) {
        $response['error'] = true;
        $response['message'] = 'Error: Both inputs must be valid numbers.';
    } else {
        $num1 = floatval($num1);
        $num2 = floatval($num2);

        switch ($operation) {
            case 'add':
                $response['result'] = $num1 + $num2;
                break;
            case 'subtract':
                $response['result'] = $num1 - $num2;
                break;
            case 'multiply':
                $response['result'] = $num1 * $num2;
                break;
            case 'divide':
                if ($num2 == 0) {
```



```
$response['error'] = true;
$response['message'] = 'Error: Cannot divide by zero.';
} else {
    $response['result'] = $num1 / $num2;
}
break;
default:
    $response['error'] = true;
    $response['message'] = 'Error: Invalid operation specified.';
    break;
}
}
} else {
    $response['error'] = true;
    $response['message'] = 'Error: Missing required parameters (num1, num2, operation).';
}
echo json_encode($response);

?>
```

Output:

Calculator

Using a PHP Web Service

First Number

Second Number

Result: 100